

Features, Labels, Targets

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the feature contract for Features labels and prediction time. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Features labels and prediction time. Proposed explanation: “Every stored column is a feature”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a new task independently

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes. Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify inputs, order, target and exclusions. For a past trip with steps 30, queue 2, arrival 10, give features and label. For a new trip with steps 24 and queue 0, give features and explain whether slow is known.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate input set.

6. Solve the given case

Rows contain leaf length, edge type and healthy/spotted label.

Identify features, target and prediction-time inputs.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two features and the target.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 2 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Define the Plant Classifier target and a first feature list; record units, allowed values, collection method, and availability time.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark

each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate input set.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Assign the roles

For the worked loan table, identify the two candidate features, the target label and the tracking field. Explain why days_to_return is not a predictor.

My reasoning and evidence

13. W2 Trace the feature contract

A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered feature list

and target. Then state what changes if it took 15 days instead.

My reasoning and evidence

14. W3 Repair a misleading score

A model has every test prediction correct after using pages and days_to_return. A learner says a separate test set makes this valid. Explain the flaw and repair the inputs.

My reasoning and evidence

15. W4 Audit a new task independently

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes. Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify inputs, order, target and exclusions. For a past trip with steps 30, queue 2, arrival 10, give features and label. For a new trip with steps 24 and queue 0, give features and explain whether slow is known.

My reasoning and evidence

16. W5 Handle an unavailable input

For the robot task, a dataset stores queue_now at export time but has no saved queue_at_dispatch. Can you rename that field to make it usable? State a defensible next step.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Features, Labels, Targets

Choose features that exist when a plant prediction must be made.

Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the target recorded later.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Features, Labels, Targets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mark prediction time

The decision occurs today, before next week's outcome.



2. Select inputs

Today's light and soil readings are available features; `plant_id` identifies the plant.



3. Keep target separate

`next_week_wilt` supplies the training label, not an input available today.

CHANGE ONE THING

A table adds `tomorrow_soil_moisture` as an input.

Explain the changed result using the matching lesson.

Why can a historical table contain a field that is an invalid input? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose features that exist when a plant prediction must be made.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? `next_week_wilt` supplies the training label, not an input available today.

1. Select inputs
2. Mark prediction time
3. Keep target separate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Keep target separate `next_week_wilt` supplies the training label, not an input available today. Read the picture in order. `next_week_wilt` supplies the training label, not an input available today. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: A table adds `tomorrow_soil_moisture` as an input. Which conclusion is supported by the supplied facts?

1. Reject it for this prediction-time contract.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject it for this prediction-time contract. Tomorrow's measurement is unavailable when today's prediction must be issued.

Repair a missing building block

Review The ML Lifecycle (ai-4/1). Goal: Distinguish fitting a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
